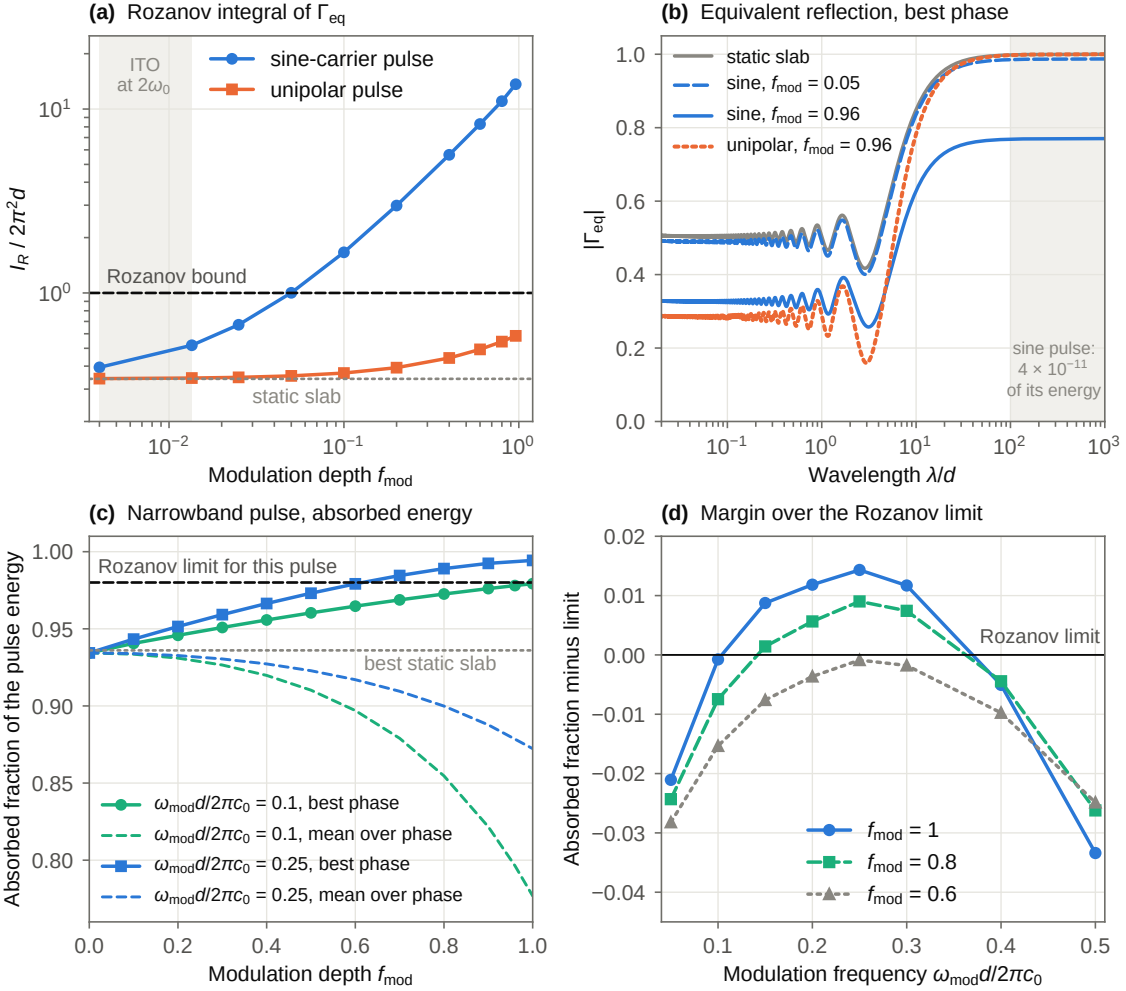
(a) Rozanov integral of Γ_{eq}
ITO at $2\omega_0$
sine-carrier pulse
unipolar pulse
Rozanov bound
static slab
$I_R / 2\pi^2 d$
Modulation depth f_{mod}
(b) Equivalent reflection, best phase
static slab
sine, f_{mod} = 0.05
sine, f_{mod} = 0.96
unipolar, f_{mod} = 0.96
sine pulse: 4×10^{-11} of its energy
$|\Gamma_{eq}|$
Wavelength λ/d
(c) Narrowband pulse, absorbed energy
Rozanov limit for this pulse
best static slab
$\omega_{mod}d/2\pi c_0$ = 0.1, best phase
$\omega_{mod}d/2\pi c_0$ = 0.1, mean over phase
$\omega_{mod}d/2\pi c_0$ = 0.25, best phase
$\omega_{mod}d/2\pi c_0$ = 0.25, mean over phase
Absorbed fraction of the pulse energy
Modulation depth f_{mod}
(d) Margin over the Rozanov limit
Rozanov limit
f_{mod} = 1
f_{mod} = 0.8
f_{mod} = 0.6
Absorbed fraction minus limit
Modulation frequency $\omega_{mod}d/2\pi c_0$